Supplementary material

Appendix A - Search Strategy

PubMed Session Results (09 Mar 2020)

#3	**#1 AND #2**	1,303
#2	"Total Quality Management"[Mesh] OR "Quality Control"[Mesh] OR "Efficiency"[Mesh] OR "Quality Improvement"[Mesh] OR "Root Cause Analysis"[Mesh] OR quality control[tiab] OR quality management[tiab] OR quality improvement*[tiab] OR process management[tiab] OR statistical process[tiab] OR process control[tiab] OR process improvement*[tiab] OR performance management[tiab] OR performance control[tiab] OR performance improvement[tiab] OR lean[tiab] OR Six Sigma[tiab] OR Sigma Metric*[tiab] OR kaizen[tiab] OR profitable[tiab] OR turnover time[tiab] OR root cause analys*[tiab] OR PIT Crew*[tiab] OR plan-do-study-act*[tiab] OR PDSA[tiab] OR plan-do-check-act*[tiab] OR PDCA[tiab] OR Operating Room throughput[tiab] OR "Operation Room throughput"[tiab] OR Operating Room turnover[tiab] OR "Operation Room turnover"[tiab]	227,005
#1	"Pulmonary Surgical Procedures"[Mesh:NoExp] OR "Pneumonectomy"[Mesh] OR pneumonectom*[tiab] OR pneumolobectom*[tiab] OR lung lobectom*[tiab] OR pulmonary lobectom*[tiab] OR ((lung*[tiab] OR pulmonar*[tiab]) AND ("Operating Rooms"[Mesh] OR "Intraoperative Period"[Mesh] OR operati*[tiab] OR operatorive[tiab] OR operatory[tiab] OR surgery[tiab] OR surgical[tiab] OR intraoperat*[tiab] OR intra-operat*[tiab]))	158,598

Embase.com Session Results (09 Mar 2020)

#4	**#3 NOT ('conference abstract'/it OR 'conference review'/it)**	1,599
#3	#1 AND #2	2,671
#2	'quality control'/de OR 'total quality management'/exp OR 'root cause analysis'/exp OR 'six sigma'/exp OR 'lean methodology'/exp OR 'lean six sigma'/exp OR 'quality control':ab,ti,kw OR 'quality management':ab,ti,kw OR 'quality improvement*':ab,ti,kw OR 'process management':ab,ti,kw OR 'statistical process':ab,ti,kw OR 'process control':ab,ti,kw OR 'process improvement*':ab,ti,kw OR 'performance management':ab,ti,kw OR 'performance control':ab,ti,kw OR 'performance improvement':ab,ti,kw OR lean:ab,ti,kw OR 'six sigma':ab,ti,kw OR 'sigma metric*':ab,ti,kw OR kaizen:ab,ti,kw OR profitable:ab,ti,kw OR 'turnover time':ab,ti,kw OR 'root cause analys*':ab,ti,kw OR 'pit crew*':ab,ti,kw OR (plan NEXT/2 do NEXT/2 study NEXT/2 act*):ab,ti,kw OR PDSA:ab,ti,kw OR (plan NEXT/2 do NEXT/2 check NEXT/2 act*):ab,ti,kw OR PDCA:ab,ti,kw OR 'Operating Room throughput':ab,ti,kw OR	363,921

	'Operation Room throughput':ab,ti,kw OR 'Operating Room turnover':ab,ti,kw OR 'Operation Room turnover':ab,ti,kw	
#1	'lung surgery'/de OR 'lung lobectomy'/exp OR pneumonectom*:ab,ti,kw OR pneumolobectom*:ab,ti,kw OR 'lung lobectom*':ab,ti,kw OR 'pulmonary lobectom*':ab,ti,kw OR ((lung*:ab,ti,kw OR pulmonar*:ab,ti,kw) AND (operati*:ab,ti,kw OR operatorive:ab,ti,kw OR operatory:ab,ti,kw OR surgery:ab,ti,kw OR surgical:ab,ti,kw OR intraoperat*:ab,ti,kw OR 'intra-operat*':ab,ti,kw))	232,110

Wiley / Cochrane Library Session Results (09 Mar 2020)

#3	**#1 AND #2**	123
#2	(quality NEXT control):ab,ti,kw OR (quality NEXT management):ab,ti,kw OR (quality NEXT improvement*):ab,ti,kw OR (process NEXT management):ab,ti,kw OR (statistical NEXT process):ab,ti,kw OR (process NEXT control):ab,ti,kw OR (process NEXT improvement*):ab,ti,kw OR (performance NEXT management):ab,ti,kw OR (performance NEXT control):ab,ti,kw OR (performance NEXT improvement):ab,ti,kw OR lean:ab,ti,kw OR (six NEXT sigma):ab,ti,kw OR (sigma NEXT metric*):ab,ti,kw OR kaizen:ab,ti,kw OR profitable:ab,ti,kw OR (turnover NEXT time):ab,ti,kw OR (root NEXT cause NEXT analys*):ab,ti,kw OR (pit NEXT crew*):ab,ti,kw OR (plan NEXT/2 do NEXT/2 study NEXT/2 act*):ab,ti,kw OR PDSA:ab,ti,kw OR (plan NEXT/2 do NEXT/2 check NEXT/2 act*):ab,ti,kw OR PDCA:ab,ti,kw OR (Operating NEXT Room NEXT throughput):ab,ti,kw OR (Operation NEXT Room NEXT throughput):ab,ti,kw OR (Operating NEXT Room NEXT turnover):ab,ti,kw OR (Operation NEXT Room NEXT turnover):ab,ti,kw	12,660
#1	pneumonectom*:ab,ti,kw OR pneumolobectom*:ab,ti,kw OR (lung NEXT lobectom*):ab,ti,kw OR (pulmonary NEXT lobectom*):ab,ti,kw OR ((lung*:ab,ti,kw OR pulmonar*:ab,ti,kw) AND (operati*:ab,ti,kw OR operatorive:ab,ti,kw OR operatory:ab,ti,kw OR surgery:ab,ti,kw OR surgical:ab,ti,kw OR intraoperat*:ab,ti,kw OR (intra NEXT operat*):ab,ti,kw))	18,241

Scopus Session Results (09 Mar 2020)

#3	**#1 AND #2**	1,249
#2	TITLE-ABS ("quality control" OR "quality management" OR "quality improvement*" OR "process management" OR "statistical process" OR "process control" OR "process improvement*" OR "performance management" OR "performance control" OR "performance improvement" OR lean OR "six sigma" OR "sigma metric*" OR kaizen OR profitable OR "turnover time" OR "root cause analys*" OR "pit crew*" OR (plan PRE/1 do PRE/1 study PRE/1 act*) OR PDSA OR (plan PRE/1 do PRE/1 check PRE/1 act*) OR PDCA OR "Operating Room throughput" OR "Operation Room throughput" OR "Operating Room turnover" OR "Operation Room turnover") OR AUTHKEY ("quality control" OR "quality management" OR "quality improvement*" OR "process management" OR "statistical process" OR "process control" OR "process improvement*" OR "performance management" OR "performance control" OR "performance improvement" OR lean OR "six sigma" OR "sigma metric*" OR kaizen OR profitable OR "turnover time" OR "root cause analys*" OR "pit crew*" OR (plan PRE/1 do PRE/1 study PRE/1 act*) OR PDSA OR (plan PRE/1 do PRE/1 check PRE/1 act*) OR PDCA OR "Operating Room throughput" OR "Operation Room throughput" OR "Operating Room turnover" OR "Operation Room turnover")	505,619
#1	TITLE-ABS (pneumonectom* OR pneumolobectom* OR "lung lobectom*" OR "pulmonary lobectom*" OR ((lung* OR pulmonar*) AND (operati* OR operatorive OR operatory OR surgery OR surgical OR intraoperat* OR "intra-operat*"))) OR AUTHKEY (pneumonectom* OR pneumolobectom* OR "lung lobectom*" OR "pulmonary lobectom*" OR ((lung* OR pulmonar*) AND (operati* OR operatorive OR operatory OR surgery OR surgical OR intraoperat* OR "intra-operat*")))	173,513

Appendix B – PRISMA Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	3
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	3
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	3
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	13-15 (Supplementary material)
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	3
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	3-4
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	3
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this	-

		information is to be used in any data synthesis.	
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5-7
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	4

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	10
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	-
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	4-5
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	6-8
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	6-8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	6-8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	-
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	-
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	-
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	8-10
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	10
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	10
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	10